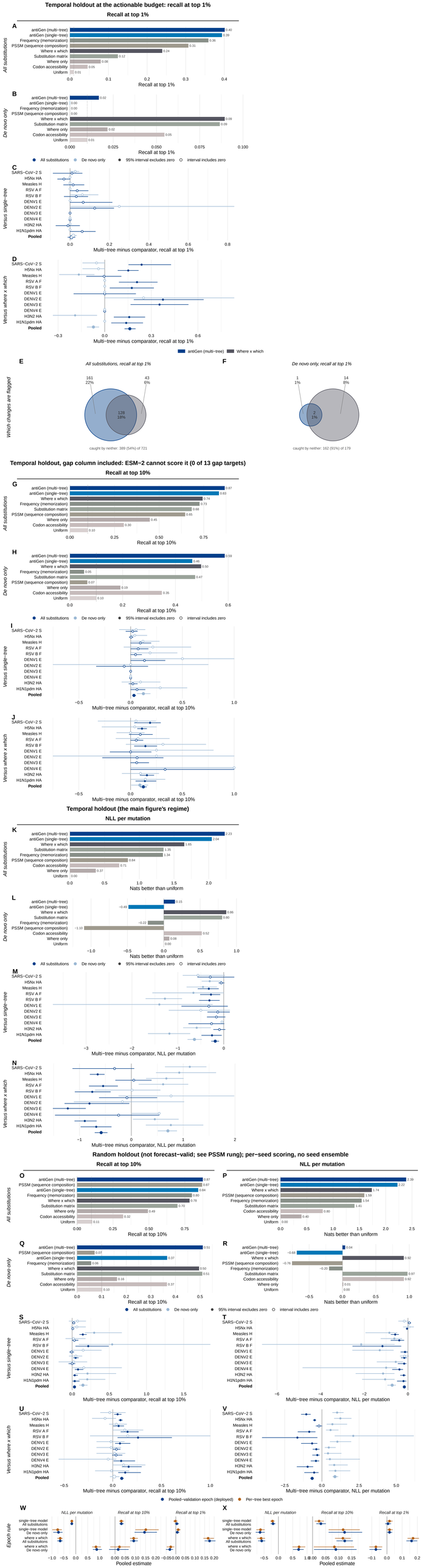
Temporal holdout at the actionable budget: recall at top 1%
Recall at top 1%
A
All substitutions
antiGen (multi−tree) 0.40
antiGen (single−tree) 0.39
Frequency (memorization) 0.36
PSSM (sequence composition) 0.31
Where x which 0.24
Substitution matrix 0.12
Where only 0.08
Codon accessibility 0.05
Uniform 0.01
0.0 0.1 0.2 0.3 0.4
Recall at top 1%
B
De novo only
antiGen (multi−tree) 0.02
antiGen (single−tree) 0.00
Frequency (memorization) 0.00
PSSM (sequence composition) 0.00
Where x which 0.09
Substitution matrix 0.09
Where only 0.02
Codon accessibility 0.05
Uniform 0.01
0.000 0.025 0.050 0.075 0.100
Recall at top 1%
All substitutions
De novo only
95% interval excludes zero
interval includes zero
C
Versus single−tree
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
0.0 0.2 0.4 0.6 0.8
Multi−tree minus comparator, recall at top 1%
D
Versus where x which
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
−0.3 0.0 0.3 0.6
Multi−tree minus comparator, recall at top 1%
antiGen (multi−tree)
Where x which
E
All substitutions, recall at top 1%
Which changes are flagged
161
22%
43
6%
128
18%
caught by neither: 389 (54%) of 721
F
De novo only, recall at top 1%
1
1%
14
8%
2
1%
caught by neither: 162 (91%) of 179
Temporal holdout, gap column included: ESM−2 cannot score it (0 of 13 gap targets)
Recall at top 10%
G
All substitutions
antiGen (multi−tree) 0.87
antiGen (single−tree) 0.83
Where x which 0.74
Frequency (memorization) 0.73
Substitution matrix 0.68
PSSM (sequence composition) 0.65
Where only 0.45
Codon accessibility 0.30
Uniform 0.10
0.00 0.25 0.50 0.75
Recall at top 10%
H
De novo only
antiGen (multi−tree) 0.59
antiGen (single−tree) 0.46
Where x which 0.50
Frequency (memorization) 0.05
Substitution matrix 0.47
PSSM (sequence composition) 0.07
Where only 0.19
Codon accessibility 0.35
Uniform 0.10
0.0 0.2 0.4 0.6
Recall at top 10%
All substitutions
De novo only
95% interval excludes zero
interval includes zero
I
Versus single−tree
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
−0.5 0.0 0.5 1.0
Multi−tree minus comparator, recall at top 10%
J
Versus where x which
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
−0.5 0.0 0.5 1.0
Multi−tree minus comparator, recall at top 10%
Temporal holdout (the main figure's regime)
NLL per mutation
K
All substitutions
antiGen (multi−tree) 2.23
antiGen (single−tree) 2.04
Where x which 1.65
Substitution matrix 1.35
Frequency (memorization) 1.34
PSSM (sequence composition) 0.84
Codon accessibility 0.71
Where only 0.37
Uniform 0.00
0.0 0.5 1.0 1.5 2.0
Nats better than uniform
L
De novo only
antiGen (multi−tree) 0.15
antiGen (single−tree) −0.49
Where x which 0.86
Substitution matrix 0.80
Frequency (memorization) −0.22
PSSM (sequence composition) −1.10
Codon accessibility 0.52
Where only 0.08
Uniform 0.00
−1.0 −0.5 0.0 0.5 1.0
Nats better than uniform
All substitutions
De novo only
95% interval excludes zero
interval includes zero
M
Versus single−tree
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
−3 −2 −1 0
Multi−tree minus comparator, NLL per mutation
N
Versus where x which
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
−1 0 1 2
Multi−tree minus comparator, NLL per mutation
Random holdout (not forecast−valid; see PSSM rung); per−seed scoring, no seed ensemble
Recall at top 10%
O
All substitutions
antiGen (multi−tree) 0.87
PSSM (sequence composition) 0.87
antiGen (single−tree) 0.84
Frequency (memorization) 0.80
Where x which 0.76
Substitution matrix 0.70
Where only 0.49
Codon accessibility 0.32
Uniform 0.11
0.00 0.25 0.50 0.75
Recall at top 10%
Q
De novo only
antiGen (multi−tree) 0.51
PSSM (sequence composition) 0.07
antiGen (single−tree) 0.37
Frequency (memorization) 0.06
Where x which 0.50
Substitution matrix 0.51
Where only 0.16
Codon accessibility 0.37
Uniform 0.10
0.0 0.1 0.2 0.3 0.4 0.5
Recall at top 10%
All substitutions
De novo only
95% interval excludes zero
interval includes zero
S
Versus single−tree
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
0.0 0.5 1.0 1.5
Multi−tree minus comparator, recall at top 10%
U
Versus where x which
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
−0.5 0.0 0.5 1.0
Multi−tree minus comparator, recall at top 10%
NLL per mutation
P
antiGen (multi−tree) 2.39
antiGen (single−tree) 2.22
Where x which 1.74
PSSM (sequence composition) 1.59
Frequency (memorization) 1.54
Substitution matrix 1.41
Codon accessibility 0.80
Where only 0.40
Uniform 0.00
0.0 0.5 1.0 1.5 2.0 2.5
Nats better than uniform
R
antiGen (multi−tree) 0.04
antiGen (single−tree) −0.69
Where x which 0.92
PSSM (sequence composition) −0.76
Frequency (memorization) −0.20
Substitution matrix 0.97
Codon accessibility 0.92
Where only 0.03
Uniform 0.00
−0.5 0.0 0.5 1.0
Nats better than uniform
T
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
−6 −4 −2 0
Multi−tree minus comparator, NLL per mutation
V
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
−2.5 0.0 2.5 5.0
Multi−tree minus comparator, NLL per mutation
Pooled−validation epoch (deployed)
Per−tree best epoch
W
Epoch rule
NLL per mutation
Recall at top 10%
Recall at top 1%
single−tree model
All substitutions
single−tree model
De novo only
where x which
All substitutions
where x which
De novo only
−1.0 −0.5 0.0 0.5 1.0
0.00 0.05 0.10 0.15 0.20
−0.05 0.00 0.05 0.10 0.15 0.20
Pooled estimate
X
NLL per mutation
Recall at top 10%
Recall at top 1%
single−tree model
All substitutions
single−tree model
De novo only
where x which
All substitutions
where x which
De novo only
−0.5 0.0 0.5 1.0
0.00 0.05 0.10 0.15
−0.1 0.0 0.1 0.2
Pooled estimate